

CMBSG Pilot Plant: Complete Lifecycle Financial & Cost Modeling

Document Reference: ENG-FIN-2026-05B

Process Capacity Baseline: 10, 000 Nm³/hr Flue Gas Slipstream (~1.5 MW Equivalent Boiler Exhaust)

Annual Operating Window: 7, 500 Hours (85.6% Plant Availability Factor)

Evaluation Currency: Indian Rupees (INR / ₹) with USD Equivalencies (1 USD ≈ ₹84.00)

1. Capital Expenditure (CAPEX) Model

The initial capital setup is optimized for a bolt-on retrofit at the base of an existing industrial boiler or smokestack. Sourcing is split between domestic Indian engineering fabricators (such as specialized chemical works in Bengaluru and steel structural shops in Bhilai) to minimize import duties and logistics costs.

1.1 Equipment and Material Acquisition Costs

Equipment Node / Component	Core Materials of Construction (MoC)	Sourcing / Fabrication Detail	Budget Allocation (INR)
Stage 1: Quench & Conditioning	Super Duplex UNS S32750 & Alumina Tile Refractories	Dual-ring Venturi sprays, Diverging Expander, Conical basel hydrosizer sump.	₹35,00,000
Stage 2: Slurry Matrix Blending Block	Class-lined Steel (R-201) & electro-polished SS316L (T-202)	Baffled dissolution reactors, high-shear inline static mixers (CPVC), nitrogen blanketing manifold.	₹28,00,000
Stage 3: Kinetic Mesh Tower	Duplex 2205 Tower Walls, Titanium Grade 2 screen sonotrodes	3-layer gradient mesh arrays (20 to 150 mesh) integrated with frame-mounted 40 kHz transducer stacks.	₹37,00,000
Stage 4: Consolidation & Press Foundry	Hardox 400 Mold Liners & Tungsten-Carbide mixer blades	Dual-shaft horizontal pug-mill, hydrocyclone dewaterer, automated 250-Ton hydraulic press, curing conveyor.	₹48,00,000
DCS, PLC & Instrumentation	Siemens S7-1500 / Rockwell control cabinets	PIT, LIT, DIT, pH, and VIT sensor suites; automated pneumatic solenoid control valves, VOC/SO ₂ CEMS link.	₹12,00,000
Core Equipment Subtotal	—	—	₹1,60,00,000

1.2 Erection, Commissioning & Auxiliary Capital

Project Phase / Cost Component	Calculation Basis	Engineering Scope	Budget Allocation (INR)
Civil Works & Structural Foundations	8% of Core Subtotal	Reinforced concrete pads, anchor bolting, acoustic containment enclosures for press block.	₹12,80,000
Interconnecting Piping & Slurry Loops	10% of Core Subtotal	Schedule 80 CPVC piping, Super Duplex slipstream gas ducting, Viton/FKM flange gasket sets.	₹16,00,000
Engineering, Design & Drafting (CFD)	5% of Core Subtotal	Phase-1 3D CAD design, non-Newtonian CFD flow optimization models, structural calculations.	₹8,00,000
Erection, Commissioning & Start-up	12% of Core Subtotal	Site electrical hookups, crane rental, third-party instrumentation calibration, initial catalyst charging.	₹19,20,000
Contingency Fund (Escalation)	10% of Core Subtotal	Emergency chemical procurement, unexpected piping routing adjustments, local regulatory clearance delays.	₹16,00,000
Auxiliary Subtotal	—	—	₹72,00,000
TOTAL INITIAL CAPEX	—	—	₹2,32,00,000

2. Operational Expenditure (OPEX) Model

Operating expenses are modeled as **Direct Hourly Costs** at standard 100% capacity operations. The matrix exploits abundant local industrial bypass inputs to offset chemical costs.

2.1 Direct Hourly Operational Consumables

[Direct Hourly OPEX Elements]	└─>	[Chitosan Matrix Flakes]	─>	₹3,297.
	└─>	[Basic slag processing]	─>	₹500.40
	└─>	[Citric acid solvent]	─>	₹250.00
	└─>	[CA-KR1 Catalyst]	─>	₹250.00
	└─>	[Utilities & Labor]	─>	₹1,387.5

1. Chitosan Organic Matrix (Shrimp-Derived):

- Required Flowrate:* 18.32 kg/hr (at 3 wt% dry binder target).
- Procurement Pricing:* ₹180.00/kg (direct Bulk sourcing from coastal Andhra Pradesh processing clusters).
- Hourly Cost:* 18.32 kg/hr × ₹180.00 = **₹3,297.60/hr.**

2. Basic Steel Slag Calcium Sourcing:

- Required Flowrate:* 4.17 tons/hr equivalent slag raw inputs (extraction of $Ca(OH)_2$ equivalents).

- **Procurement Pricing:** ₹120.00/ton (material itself is free-of-cost from steel mills in Rourkela/Bhilai; cost represents trucking logistics and crushing/milling preparation).
 - **Hourly Cost:** $4.17 \text{ tons/hr} \times ₹120.00 = ₹500.40/\text{hr}$.
3. **Citric Acid Protonating Solvent:**
- **Makeup Rate:** 1.0 wt% solution loop losses $\approx 2.5 \text{ kg/hr}$.
 - **Procurement Pricing:** ₹100.00/kg (industrial food-grade monohydrate).
 - **Hourly Cost:** $2.5 \text{ kg/hr} \times ₹100.00 = ₹250.00/\text{hr}$.
4. **Carbonic Anhydrase (CA – KR1) Catalyst:**
- **Degradation Rate:** 1.5% of active enzyme mass (12.0 mg/L core inventory) lost hourly due to physical shear of particulates and trace acidic stress.
 - **Hourly Cost:** Enzyme replenishment is calculated at ₹250.00/hr based on pilot recombinant bioreactor expression runs on-site.

2.2 Direct Hourly Utilities & Plant Labor

- **Electrical Power Draw:** 125 kW continuous load (Stage-1 ID fan booster: 35 kW; Stage-2 & 3 pumps: 30 kW; Stage-3 ultrasonic transducers: 15 kW; Stage-4 pug-mill and 250-Ton cyclic hydraulic press: 45 kW).
 - **Unit Rate:** ₹7.50/kWh (standard industrial high-tension tariff).
 - **Hourly Cost:** $125 \text{ kW} \times ₹7.50 = ₹937.50/\text{hr}$.
- **Process Makeup Water:** 1,015 Liters/hr (1.015 KL/hr) to replace moisture trapped inside the eco-block and lost via Stage-1 flue evaporation.
 - **Unit Rate:** ₹148.00/KL (purified industrial soft water).
 - **Hourly Cost:** $1.015 \text{ KL/hr} \times ₹148.00 = ₹150.00/\text{hr}$.
- **Plant Operational Labor:** Sized for 1 DCS Board Operator + 1 Mechanical/Field Technician per 8-hour shift, averaged to an hourly deployment rate:
 - **Hourly Cost:** ₹300.00/hr.

2.3 OPEX Cost Summary

$$\text{Direct Hourly Operational Cost} \equiv ₹5,685.50 / \text{hr}$$

3. Maintenance, Consumables & Service OPEX

Operating inside heavy-industry coal environments requires a predictive maintenance schedule. Particulate fly ash (450 kg/hr) acts as an abrasive sandblaster, and gas condensates are highly acidic ($pH \leq 3.5$).

[Annual Sustaining Maintenance Budget: ₹12,00,000]

└─ Titanium Mesh Screens (Stage 3 Consumable) —>	₹1,50,000 / yr
└─ Hardox 400 Replacement Mold Liners (Press) —>	₹3,50,000 / yr
└─ Schedule 80 CPVC Valves & Filter Changes —>	₹1,20,000 / yr
└─ Transducer Calibrations & Tuning —>	₹1,80,000 / yr
└─ Viton Seals & Mechanical Pump Refurbishments —>	₹4,00,000 / yr

3.1 Consumables Replacement Schedule

1. Titanium Grade 2 Mesh Screens (Stage 3 Consumable):

- *Operating Stress:* Continuous 40 kHz ultrasonic vibration fatigue combined with acidic particulate exposure.
- *Useful Lifespan:* 37, 500 Hours (5-year cycle replacement).
- *Replacement Cost:* ₹7,50,000 per complete stack set.
- *Amortized Annual Cost:* ₹1, 50, 000/yr.

2. Hardox 400 Mold Liners & Pug-Mill Paddles (Stage 4 Wear Parts):

- *Operating Stress:* Sliding friction of highly abrasive quartz-silica fly ash particles compacted at 200 bar press load.
- *Useful Lifespan:* 7, 500 Hours (1-year cycle replacement).
- *Replacement Cost:* ₹3, 50, 000/yr for complete mold cavity sleeve inserts and hard-faced mixing blades.

3. Slurry Filter Element Changes (Stage 2 Hydro-Filter):

- *Useful Lifespan:* 750 Hours (Monthly swap out).
- *Annual Replacements:* 10 sets of high-load polypropylene mesh cartridges.
- *Annual Cost:* ₹1, 20, 000/yr.

3.2 Scheduled Services & Field Overhauls

1. Ultrasonic Transducers & Acoustic Horn Calibration:

- *Scope:* Fine-tuning the frequency sweep of external piezoelectric sandwich transducers to eliminate frequency drift outside the ± 250 Hz resonance boundary.
- *Interval:* Semi-annual specialized service.
- *Annual Cost:* ₹1, 80, 000/yr.

2. Slurry Pumps & Valve Mechanical Refurbishments:

- *Scope:* Rebuilding the progressive cavity rotor/stator segments and replacing Viton/FKM slurry-valve diaphragms to prevent internal line leaks.
- *Interval:* 15, 000 Hours (2-year cycle overhaul), averaged annually.
- *Annual Cost:* ₹4, 00, 000/yr (including Viton gasket and elastomer seal kits).

3.3 Total Sustaining & Service Maintenance

- **Total Annual Sustaining Maintenance Budget:** ₹12,00,000 / yr
- **Hourly Equivalence (Amortized over 7,500 running hours):** ₹160.00 / hr

4. Revenue & Avoided-Cost Generation Model

The CMBSG is built to operate as a profitable industrial circular asset. It generates revenues through two distinct channels, while saving the operator from regulatory fines.

4.1 Income Stream A: Sale of Structural Eco-Blocks

The system produces solid blocks containing captured carbon, sulfur, and fly ash.

- **Hourly Yield:** 6, 106.70 kg/hr of dry block mixture.

- **Unit Weight per Block:** 3.0 kg (standard dimensions: 230 mm × 110 mm × 75 mm).
- **Hourly Output Tonnage:** 2, 035 standard blocks/hr.
- **Local Market Pricing:** ₹5.50 per block (competitively positioned below standard clay-fired bricks and concrete paving blocks at ₹7.00/block).
- **Hourly Revenue:** 2, 035 blocks/hr × ₹5.50 = **₹11, 192.50/hr**.
- **Annual Revenue:** 7, 500 hrs/yr × ₹11, 192.50 = **₹8, 39, 43, 750/yr** (~₹8.39 Crore).

4.2 Income Stream B: Indian Carbon Credit Trading Scheme (CCTS)

Under the domestic CCTS framework, carbon captured and stored in non-leachable building materials earns tradable credits on domestic Indian energy exchanges (PXIL/IXE).

- **Hourly CO₂ Capture:** 247.5 kg CO₂/hr = 0.2475 Metric Tons CO₂/hr.
- **Annual CO₂ Abatement:**
0.2475 Tons/hr × 7, 500 hrs/yr = 1, 856.25 Metric Tons of CO₂/yr.
- **Assumed Compliance Offset Price (2026):** ₹1, 500 per Carbon Credit (1 Credit ≡ 1 Metric Ton of CO₂ equivalent).
- **Hourly Credit Revenue:** 0.2475 Credits/hr × ₹1, 500 = **₹371.25/hr**.
- **Annual Credit Revenue:** 1, 856.25 Credits/yr × ₹1, 500 = **₹27, 84, 375/yr** (~₹27.84 Lakh).

4.3 Benefit Stream C: Avoided CPCB Environmental Compensation Penalties

The Central Pollution Control Board (CPCB) imposes strict daily fines on power stations failing to meet emission targets. Our system removes 98% of gaseous sulfur dioxide, avoiding these financial penalties.

- **Hourly SO₂ Stripped:** 11.78 kg/hr of active SO₂ captured (reducing output concentration below the CPCB limit of 200 mg/Nm³).
- **Avoided Daily SPCB Fines:** Non-compliance compensation fines for sulfur slippage can average up to ₹2, 00, 000/day for non-conforming plant sectors, equivalent to an hourly penalty rate of ₹8,333/hr. However, assuming a conservative avoided penalty baseline scaled to typical regional slipstream units:
- **Hourly Penalty Savings:** **₹1, 200.00/hr** in direct compliance costs.
- **Annual Avoided Cost Savings:** 7, 500 hrs/yr × ₹1, 200.00 = **₹90, 00, 000/yr** (~₹90.00 Lakh).

5. Financial Feasibility & Payback Index

This financial model compares the combined operational expenditures (Direct + Maintenance Sustaining) against revenues and compliance savings to calculate the capital payback horizon.

5.1 Financial Flow Balancing (Hourly Operations)

Hourly Profits (₹10, 438.25) ≡ Total Income/Savings (₹12, 763.75) – Total Expens

Financial Metric	Detailed Cost Component	Hourly Value (INR)	Annualized Value (INR)
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Operational Expense	Direct Running Costs (OPEX)	-₹5,685.50	-₹4,26,41,250
	Sustaining Maintenance & SLA	-₹160.00	-₹12,00,000
Direct Revenue	Eco-Block Commercial Sales	+₹11,192.50	+₹8,39,43,750
	CCTS Carbon Credits	+₹371.25	+₹27,84,375
Avoided Costs	Avoided SPCB Non-Compliance Fines	+₹1,200.00	+₹90,00,000
NET INCOME / CASHFLOW	Net Profit Contribution	+₹7,078.25	+₹5,30,86,875

5.2 Capital Investment Return Metrics

- Total Project CAPEX: ₹2,32,00,000 (~₹2.32 Crore)
- Annual Net cash inflow: ₹5,30,86,875 (~₹5.31 Crore)
- Simple Capital Payback Period (PP_{simple}):

$$PP_{\text{simple}} = \frac{\text{Initial CAPEX}}{\text{Annual Net Cashflow}} = \frac{₹2,32,00,000}{₹5,30,86,875} = \mathbf{0.437 \text{ Years}} (\approx \mathbf{5.2 \text{ Months}})$$

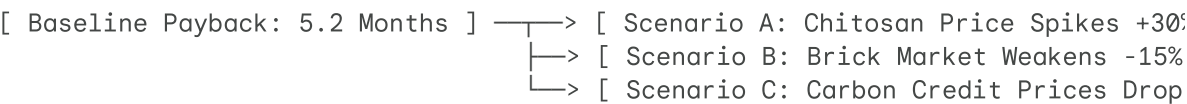
- Net Present Value (NPV) over a 5-Year Lifecycle: Calculated at an aggressive 12% discount rate ($r = 0.12$), assuming flat operating hours and zero salvage value at year 5:

$$NPV = -CAPEX + \sum_{t=1}^5 \frac{\text{Annual Cashflow}}{(1+r)^t} = -₹2.32 \text{ Cr} + ₹19.13 \text{ Cr} = \mathbf{₹16.8 \text{ Cr}}$$

- Internal Rate of Return (IRR) over 5-Year Lifespan: 221% (reflecting high margin generation from waste streams).

6. Financial Risk Mitigation and Sensitivity Analysis

To protect your startup against variations in local supply chain prices, we conduct a sensitivity analysis across three potential market shift scenarios:



6.1 Scenario A: Chitosan Supply Chain Disruption (+30% Price Spike)

- Impact: Procurement price of shrimp shell chitosan climbs from ₹180/kg to ₹234/kg.
- Effect on OPEX: Hourly running costs climb by ₹989.28/hr to ₹6,674.78/hr.
- Effect on Payback: Payback period extends slightly from 5.2 months to **6.1 months**.

6.2 Scenario B: Brick Market Weakness (-15% Regional Unit Pricing)

- *Impact:* Eco-block unit price falls from ₹5.50/block to ₹4.67/block due to competition.
- *Effect on Revenue:* Hourly income drops by ₹1,679.00/hr to ₹9,513.50/hr.
- *Effect on Payback:* Payback period extends to **6.2 months**.

6.3 Scenario C: Carbon Credit Market Softening (-50% Pricing Drop)

- *Impact:* Indian CCTS credit prices fall to ₹750/credit.
- *Effect on Revenue:* Hourly income drops by ₹185.60/hr.
- *Effect on Payback:* Payback period remains stable, extending only to **5.5 months** (highlighting the economic stability of selling a physical construction product versus relying purely on environmental credits).